

Bilateral Filtering for Gray and Color Images

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Abstract

Bilateral filtering smooths images while preserving edges, by means of a nonlinear combination of nearby image values. The method is noniterative, local, and simple. It combines gray levels or colors based on both their geometric closeness and their photometric similarity, and prefers near values to distant values in both domain and range. In contrast with filters that operate on the three bands of a color image separately, a bilateral filter can enforce the perceptual metric underlying the CIE-Lab color space, and smooth colors and preserve edges in a way that is tuned to human perception. Also, in contrast with standard filtering, bilateral filtering produces no phantom colors along edges in color images, and reduces phantom colors where they appear in the original image.

1 Introduction

Filtering is perhaps the most fundamental operation of image processing and computer vision. In the broadest sense of the term “filtering,” the value of the filtered image at a given location is a function of the values of the input image in a small neighborhood of the same location. In particular, Gaussian low-pass filtering computes a weighted average of pixel values in the neighborhood, in which, the weights decrease with distance from the neighborhood center. Although formal and quantitative explanations of this weight fall-off can be given [11], the intuition is that images typically vary slowly over space, so near pixels are likely to have similar values, and it is therefore appropriate to average them together. The noise values that corrupt these nearby pixels are mutually less correlated than the signal values, so noise is averaged away while signal is preserved.

The assumption of slow spatial variations fails at edges, which are consequently blurred by low-pass filtering. Many efforts have been devoted to reducing this undesired effect [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 17]. How can

we prevent averaging across edges, while still averaging within smooth regions? Anisotropic diffusion [12, 14] is a popular answer: local image variation is measured at every point, and pixel values are averaged from neighborhoods whose size and shape depend on local variation. Diffusion methods average over extended regions by solving partial differential equations, and are therefore inherently iterative. Iteration may raise issues of stability and, depending on the computational architecture, efficiency. Other approaches are reviewed in section 6.

In this paper, we propose a noniterative scheme for edge preserving smoothing that is noniterative and simple. Although we claim no correlation with neurophysiological observations, we point out that our scheme could be implemented by a single layer of neuron-like devices that perform their operation once per image.

Furthermore, our scheme allows explicit enforcement of any desired notion of photometric distance. This is particularly important for filtering color images. If the three bands of color images are filtered separately from one another, colors are corrupted close to image edges. In fact, different bands have different levels of contrast, and they are smoothed differently. Separate smoothing perturbs the balance of colors, and unexpected color combinations appear. Bilateral filters, on the other hand, can operate on the three bands at once, and can be told explicitly, so to speak, which colors are similar and which are not. Only perceptually similar colors are then averaged together, and the artifacts mentioned above disappear.

The idea underlying bilateral filtering is to do in the range of an image what traditional filters do in its domain. Two pixels can be *close* to one another, that is, occupy nearby spatial location, or they can be *similar* to one another, that is, have nearby values, possibly in a perceptually meaningful fashion. Closeness refers to vicinity in the domain, similarity to vicinity in the range. Traditional filtering is domain filtering, and enforces closeness by weighing pixel values with coefficients that fall off with distance. Similarly, we define range filtering, which averages image

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双边滤波在灰度和彩色图像中的应用

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摘要

Bilateral filtering smooths images while preserving edges, by means of a nonlinear combination of nearby image values. The method is noniterative, local, and simple. It combines gray levels or colors based on both their geometric closeness and their photometric similarity, and prefers near values to distant values in both domain and range. In contrast with filters that operate on the three bands of a color image separately, a bilateral filter can enforce the perceptual metric underlying the CIE-Lab color space, and smooth colors and preserve edges in a way that is tuned to human perception. Also, in contrast with standard filtering, bilateral filtering produces no phantom colors along edges in color images, and reduces phantom colors where they appear in the original image.

1 引言

图像处理和计算机视觉中最基本的操作之一是滤波。从“滤波”这一术语最广泛的意义来看,滤波后图像在某一位置的值取决于输入图像在同一位置附近小区域内的像素值。特别是高斯低通滤波计算了该邻域内像素值的加权平均值,在此过程中,权重随着距离中心点的距离增加而减小。虽然可以给出这种权重衰减形式化的定量解释[11],但直观上来说,图像通常在空间上的变化较为缓慢,因此附近的像素很可能具有相似的值,将它们进行平均是合理的。这些附近像素中的噪声值相互之间的相关性较低,信号值则相反,所以通过平均可以消除噪声而保留信号。

假设空间变化缓慢的条件在边缘处失效,导致低通滤波会模糊这些边缘。许多研究致力于减少这种不希望出现的影响[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 17]。如何才能

我们如何在使边缘平滑的情况下,仍然对平滑区域内的像素值进行平均处理?各向异性扩散[12, 14]是一种流行的方法:它在每个点测量局部图像变化,并根据局部变化的大小和形状从邻域中计算像素值的平均值。扩散方法通过求解偏微分方程来对扩展区域进行平均处理,因此本质上是迭代的。迭代可能会引发稳定性问题,并且根据计算架构的不同,还可能影响效率。其他方法将在第6节中进行回顾。

在本文中,我们提出了一种非迭代的边缘保持平滑方案,该方案简单且无需迭代。尽管我们不声称与神经生理学观察有任何关联,但我们指出,我们的方案可以通过一层类似神经元的设备实现,这些设备对每幅图像执行一次操作。

此外,我们的方案允许明确实施任何所需的光度距离概念。这在过滤彩色图像时尤为重要。如果对彩色图像的三个通道分别进行滤波处理,在接近图像边缘处颜色会受到破坏。实际上,不同的通道具有不同的对比度,并且它们被平滑的方式也不同。单独的平滑操作会扰乱颜色平衡,导致出现意外的颜色组合。另一方面,双边滤波器可以同时作用于这三个通道,并且能够明确指定哪些颜色相似以及哪些不相似。只有在感知上相似的颜色才会被平均处理,上述提到的伪影也会随之消失。

双边滤波的基本思想是在图像的值域中执行类似于传统滤波在空间域中的操作。两个像素可以是**邻近位置**,即它们占据相近的空间位置;或者它们可以是**相似值**,即它们具有接近的数值,这种接近可能是感知上有意义的方式。这里的“邻近”指的是空间域上的接近,“相似”则指值域上的接近。传统滤波是在空间域上进行的操作,并通过随着距离增加而衰减的权重系数来强制执行像素之间的邻近性。类似地,我们定义了值域滤波,它通过对图像中的像素值进行平均来实现这一点。

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values with weights that decay with dissimilarity. Range filters are nonlinear because their weights depend on image intensity or color. Computationally, they are no more complex than standard nonseparable filters. Most importantly, they preserve edges, as we show in section 4.

Spatial locality is still an essential notion. In fact, we show that range filtering by itself merely distorts an image's color map. We then combine range and domain filtering, and show that the combination is much more interesting. We denote the combined filtering as *bilateral* filtering.

Since bilateral filters assume an explicit notion of distance in the domain and in the range of the image function, they can be applied to any function for which these two distances can be defined. In particular, bilateral filters can be applied to color images just as easily as they are applied to black-and-white ones. The CIE-Lab color space [16] endows the space of colors with a perceptually meaningful measure of color similarity, in which short Euclidean distances correlate strongly with human color discrimination performance [16]. Thus, if we use this metric in our bilateral filter, images are smoothed and edges are preserved in a way that is tuned to human performance. Only perceptually similar colors are averaged together, and only perceptually visible edges are preserved.

In the following section, we formalize the notion of bilateral filtering. Section 3 analyzes range filtering in isolation. Sections 4 and 5 show experiments for black-and-white and color images, respectively. Relations with previous work are discussed in section 6, and ideas for further exploration are summarized in section 7.

2 The Idea

A low-pass domain filter applied to image $\mathbf{f}(\mathbf{x})$ produces an output image defined as follows:

$$\mathbf{h}(\mathbf{x}) = k_d^{-1}(\mathbf{x}) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{f}(\xi) c(\xi, \mathbf{x}) d\xi \quad (1)$$

where $c(\xi, \mathbf{x})$ measures the *geometric* closeness between the neighborhood center $\mathbf{x}$ and a nearby point ξ . The bold font for $\mathbf{f}$ and $\mathbf{h}$ emphasizes the fact that both input and output images may be multiband. If low-pass filtering is to preserve the dc component of low-pass signals we obtain

$$k_d(\mathbf{x}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c(\xi, \mathbf{x}) d\xi \quad (2)$$

If the filter is shift-invariant, $c(\xi, \mathbf{x})$ is only a function of the vector difference $\xi - \mathbf{x}$, and k_d is constant.

Range filtering is similarly defined:

$$\mathbf{h}(\mathbf{x}) = k_r^{-1}(\mathbf{x}) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{f}(\xi) s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x})) d\xi \quad (3)$$

except that now $s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x}))$ measures the *photometric* similarity between the pixel at the neighborhood center $\mathbf{x}$ and

that of a nearby point ξ . Thus, the similarity function s operates in the range of the image function $\mathbf{f}$, while the closeness function c operates in the domain of $\mathbf{f}$. The normalization constant (2) is replaced by

$$k_r(\mathbf{x}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x})) d\xi \quad (4)$$

Contrary to what occurs with the closeness function c , the normalization for the similarity function s depends on the image $\mathbf{f}$. We say that the similarity function s is *unbiased* if it depends only on the difference $\mathbf{f}(\xi) - \mathbf{f}(\mathbf{x})$.

The spatial distribution of image intensities plays no role in range filtering taken by itself. Combining intensities from the entire image, however, makes little sense, since image values far away from $\mathbf{x}$ ought not to affect the final value at $\mathbf{x}$. In addition, section 3 shows that range filtering by itself merely changes the color map of an image, and is therefore of little use. The appropriate solution is to combine domain and range filtering, thereby enforcing both geometric and photometric locality. Combined filtering can be described as follows:

$$\mathbf{h}(\mathbf{x}) = k^{-1}(\mathbf{x}) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{f}(\xi) c(\xi, \mathbf{x}) s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x})) d\xi \quad (5)$$

with the normalization

$$k(\mathbf{x}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c(\xi, \mathbf{x}) s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x})) d\xi \quad (6)$$

Combined domain and range filtering will be denoted as *bilateral filtering*. It replaces the pixel value at $\mathbf{x}$ with an average of similar and nearby pixel values. In smooth regions, pixel values in a small neighborhood are similar to each other, and the normalized similarity function $k^{-1}s$ is close to one. As a consequence, the bilateral filter acts essentially as a standard domain filter, and averages away the small, weakly correlated differences between pixel values caused by noise. Consider now a sharp boundary between a dark and a bright region, as in figure 1 (a). When the bilateral filter is centered, say, on a pixel on the bright side of the boundary, the similarity function s assumes values close to one for pixels on the same side, and close to zero for pixels on the dark side. The similarity function is shown in figure 1 (b) for a 23×23 filter support centered two pixels to the right of the step in figure 1 (a). The normalization term $k(\mathbf{x})$ ensures that the weights for all the pixels add up to one. As a result, the filter replaces the bright pixel at the center by an average of the bright pixels in its vicinity, and essentially ignores the dark pixels. Conversely, when the filter is centered on a dark pixel, the bright pixels are ignored instead. Thus, as shown in figure 1 (c), good filtering behavior is achieved at the boundaries, thanks to the domain component of the filter, and crisp edges are preserved at the same time, thanks to the range component.

具有权重的值，这些权重随不相似性而衰减。范围滤波器是非线性的，因为它们的权重依赖于图像的强度或颜色。从计算复杂度来看，它们并不比标准的非可分离滤波器更复杂。最重要的是，如第4节所示，它们能够保持边缘信息。

空间局部性仍然是一个基本概念。事实上，我们证明了范围过滤本身只会扭曲图像的颜色映射。然后将范围过滤与域过滤结合，并展示了这种组合更加有趣。我们将这种结合后的过滤方法表示为 *bilateral* 过滤。

由于双边滤波器假设了图像函数定义域和值域中的显式距离概念，因此只要这两个距离可以被定义，它们就可以应用于任何函数。特别是，双边滤波器同样容易地应用于彩色图像，就像应用到黑白图像一样。CIE-Lab颜色空间[16]赋予色彩空间一种感知上具有意义的颜色相似度量，在这种度量下，短的欧几里得距离与人类对颜色的辨别性能密切相关[16]。因此，如果我们在这个双边滤波器中使用这个度量，图像会被平滑处理，并且边缘会以符合人类表现的方式被保留下来。只有感知上相似的颜色才会被平均化，而只有在视觉上可见的边缘才会被保留。

在接下来的部分中，我们将形式化双边滤波的概念。第三部分单独分析范围滤波。第四部分和第五部分分别展示了黑白图像和彩色图像的实验结果。第六部分讨论了与先前工作的关系，第七部分总结了进一步探索的想法。

2 设想

低通域滤波器应用于图像 $f(x)$ ，产生的输出图像定义如下：

$$h(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\xi) c(\xi, x) q\xi \quad (1)$$

其中， $c(\xi, x)$ 表示邻域中心 x 和附近点 ξ 之间的 *geometric* 接近程度。 f 和 h 的加粗字体强调了输入和输出图像都可能是多波段的这一事实。如果低通滤波要保持低通信号中的直流分量，则可以得到

$$q_q(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c(\xi, x) q\xi \quad (2)$$

如果滤波器是移不变的，则 $c(\xi, x)$ 只是向量差 $\xi - x$ 的函数，且 q_q 是常数。

范围过滤同样定义为：

$$h(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\xi) v(f(\xi), f(x)) q\xi \quad (3)$$

除了现在 $v(f(\xi), f(x))$ 测量了以中心像素 x 为中心的邻域内的相似度之外。

与邻近点 ξ 的情况类似。因此，相似性函数 v 在图像函数 f 的值域内运行，而接近度函数 c 则在 f 的定义域内运行。(2) 中的标准化常数被替换为

$$q_v(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v(f(\xi), f(x)) q\xi \quad (4)$$

与接近度函数 c 不同，相似性函数 v 的规范化依赖于图像 f 。我们说相似性函数 v 是 *unbiased*，如果它仅取决于差值 $f(\xi) - f(x)$ 。

图像强度的空间分布本身在范围滤波中不起作用。然而，结合整个图像的强度却没有意义，因为远离点 x 的图像值不应影响最终在 x 处的值。此外，第3节表明，仅靠范围滤波只会改变图像的颜色映射表，因此用途不大。适当的解决方案是将域滤波和范围滤波结合起来，从而同时保证几何局部性和光度局部性。结合滤波可以描述如下：

$$h(x) = q_q^{-1}(x) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\xi) c(\xi, x) v(f(\xi), f(x)) q\xi \quad (5)$$

经过规范化后

$$q_v(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} c(\xi, x) v(f(\xi), f(x)) q\xi \quad (6)$$

结合域和范围的过滤将表示为 *bilateral filtering*。它用相似且邻近像素值的平均值替换位置 x 处的像素值。在平滑区域，小邻域内的像素值彼此相似，归一化相似函数 q_v^{-1} 接近于 1。因此，双边滤波器基本上作为一个标准域滤波器工作，并通过平均消除由噪声引起的微弱相关差异。现在考虑图 1 中的暗区和亮区之间的锐利边界。当双边滤波器位于边界亮侧的一个像素上时，相似函数 v 对于同一侧的像素值接近于 1，而对于另一侧的像素值则接近于 0。如图 1 (b) 所示，对于一个以图 1 (a) 中阶梯右方两个像素为中心的支持滤波器，相似函数被展示出来。归一化项 $q_v^{-1}(x)$ 确保所有像素的权重加起来为 1。因此，该滤波器用其邻近区域内的亮像素平均值替换中心处的亮像素，并基本上忽略暗像素。相反地，当滤波器位于一个暗像素上时，则会忽略亮像素。因此，如图 1 (c) 所示，在边界处实现了良好的过滤效果，这得益于滤波器的域部分，同时由于范围部分的作用，锐利边缘得以保留。

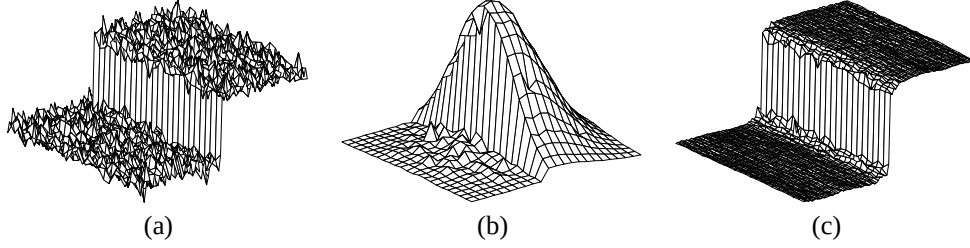


Figure 1: (a) A 100-gray-level step perturbed by Gaussian noise with $\sigma = 10$ gray levels. (b) Combined similarity weights $c(\xi, \mathbf{x})s(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x}))$ for a 23×23 neighborhood centered two pixels to the right of the step in (a). The range component effectively suppresses the pixels on the dark side. (c) The step in (a) after bilateral filtering with $\sigma_r = 50$ gray levels and $\sigma_d = 5$ pixels.

2.1 Example: the Gaussian Case

A simple and important case of bilateral filtering is shift-invariant Gaussian filtering, in which both the closeness function $c(\xi, \mathbf{x})$ and the similarity function $s(\phi, \mathbf{f})$ are Gaussian functions of the Euclidean distance between their arguments. More specifically, c is radially symmetric

$$c(\xi, \mathbf{x}) = e^{-\frac{1}{2}\left(\frac{d(\xi, \mathbf{x})}{\sigma_d}\right)^2}$$

where

$$d(\xi, \mathbf{x}) = d(\xi - \mathbf{x}) = \|\xi - \mathbf{x}\|$$

is the Euclidean distance between ξ and $\mathbf{x}$. The similarity function s is perfectly analogous to c :

$$s(\xi, \mathbf{x}) = e^{-\frac{1}{2}\left(\frac{\delta(\mathbf{f}(\xi), \mathbf{f}(\mathbf{x}))}{\sigma_r}\right)^2}$$

where

$$\delta(\phi, \mathbf{f}) = \delta(\phi - \mathbf{f}) = \|\phi - \mathbf{f}\|$$

is a suitable measure of distance between the two intensity values ϕ and $\mathbf{f}$. In the scalar case, this may be simply the absolute difference of the pixel difference or, since noise increases with image intensity, an intensity-dependent version of it. A particularly interesting example for the vector case is given in section 5.

The geometric spread σ_d in the domain is chosen based on the desired amount of low-pass filtering. A large σ_d blurs more, that is, it combines values from more distant image locations. Also, if an image is scaled up or down, σ_d must be adjusted accordingly in order to obtain equivalent results. Similarly, the photometric spread σ_r in the image range is set to achieve the desired amount of combination of pixel values. Loosely speaking, pixels with values much closer to each other than σ_r are mixed together and values much more distant than σ_r are not. If the image is amplified or attenuated, σ_r must be adjusted accordingly in order to leave the results unchanged.

Just as this form of domain filtering is shift-invariant, the Gaussian range filter introduced above is insensitive to overall additive changes of image intensity, and is therefore

unbiased: if filtering $\mathbf{f}(\mathbf{x})$ produces $\mathbf{h}(\mathbf{x})$, then the same filter applied to $\mathbf{f}(\mathbf{x}) + \mathbf{a}$ yields $\mathbf{h}(\mathbf{x}) + \mathbf{a}$, since $\delta(\mathbf{f}(\xi) + \mathbf{a}, \mathbf{f}(\mathbf{x}) + \mathbf{a}) = \delta(\mathbf{f}(\xi) + \mathbf{a} - (\mathbf{f}(\mathbf{x}) + \mathbf{a})) = \delta(\mathbf{f}(\xi) - \mathbf{f}(\mathbf{x}))$. Of course, the range filter is shift-invariant as well, as can be easily verified from expressions (3) and (4).

3 Range Versus Bilateral Filtering

In the previous section we combined range filtering with domain filtering to produce bilateral filters. We now show that this combination is essential. For notational simplicity, we limit our discussion to black-and-white images, but analogous results apply to multiband images as well. The main point of this section is that range filtering by itself merely modifies the gray map of the image it is applied to. This is a direct consequence of the fact that a range filter has no notion of space.

Let $\nu(\phi)$ be the frequency distribution of gray levels in the input image. In the discrete case, $\nu(\phi)$ is the gray level histogram: ϕ is typically an integer between 0 and 255, and $\nu(\phi)$ is the fraction of image pixels that have a gray value of ϕ . In the continuous case, $\nu(\phi) d\phi$ is the fraction of image area whose gray values are between ϕ and $\phi + d\phi$. For notational consistency, we continue our discussion in the continuous case, as in the previous section.

Simple manipulation, omitted for lack of space, shows that expressions (3) and (4) for the range filter can be combined into the following:

$$h = \int_0^\infty \phi \tau(\phi, f) d\phi \quad (7)$$

where

$$\tau(\phi, f) = \frac{s(\phi, f) \nu(\phi)}{\int_0^\infty s(\phi, f) \nu(\phi) d\phi}$$

independently of the position $\mathbf{x}$. Equation (7) shows range filtering to be a simple transformation of gray levels. The mapping kernel $\tau(\phi, f)$ is a density function, in the sense that it is nonnegative and has unit integral. It is equal to the histogram $\nu(\phi)$ weighted by the similarity function s centered at f and normalized to unit area. Since τ is

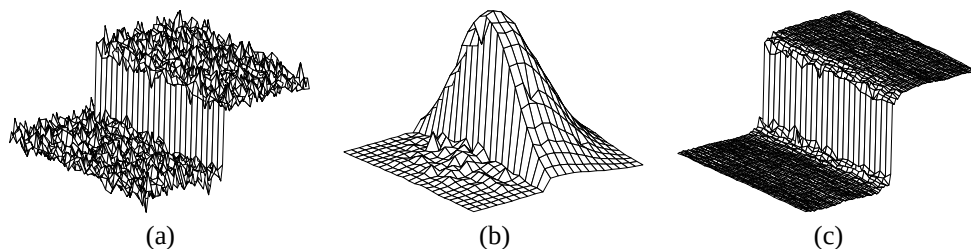


图1: (a) 一个由高斯噪声扰动的100级灰度阶梯, 包含 100 个灰度级别。(b) 针对(a)中阶梯右侧两个像素位置为中心的 33×33 邻域组合相似权重 $c(\xi, x, f(\xi), f(x))$ 。范围成分有效抑制了暗侧的像素。(c) 经双边滤波处理, 灰度级别为 $a_{\xi} = 20$ 且像素数为 $a_{\phi} = 2$ 的(a)中的阶梯。

2.1 示例: 高斯情形

双边滤波的一个简单而重要的特例是移不变高斯滤波, 在此情况下, 接近度函数 $c(\xi, x)$ 和相似性函数 $s(\phi, f)$ 均为它们的参数之间欧几里得距离的高斯函数。更具体地说, c 是径向对称的。

$$c(\xi, \mathbf{x}) = e^{-\frac{1}{2} \frac{q(\xi, \mathbf{x})^2}{a_{\phi}}}$$

其中

$$q(\xi, \mathbf{x}) = q(\xi - \mathbf{x}) = \|\xi - \mathbf{x}\|$$

是向量 ξ 和 $\mathbf{x}$ 之间的欧几里得距离。相似性函数 s 完全类似于 c :

$$s(\xi, \mathbf{x}) = e^{-\frac{1}{2} \frac{q(\xi, \mathbf{x})^2}{a_{\phi}}}$$

其中

$$q(\phi, \mathbf{f}) = q(\phi - \mathbf{f}) = \|\phi - \mathbf{f}\|$$

ϕ 和 $\mathbf{f}$ 这两个强度值之间的距离的一个合适度量。在标量情况下, 这可以是像素差的绝对值, 或者由于噪声随图像强度增加, 也可以是一个与强度相关的版本。向量情况下的一个特别有趣的例子见第5节。

在域中选择几何扩散, 基于所需的低通滤波量。较大的 a_{ϕ} 会模糊更多, 也就是说, 它结合了来自更远图像位置的值。同样, 如果图像放大或缩小, 则必须相应调整 a_{ϕ} 以获得等效结果。类似地, 在图像范围中设置光度扩散, 以实现所需的像素值组合量。粗略地说, 彼此值非常接近于 a_{ϕ} 的像素会被混合在一起, 而距离远大于 a_{ϕ} 的值则不会被结合。如果图像被放大或衰减, 则必须相应调整 a_{ϕ} 以保持结果不变。

正如这种域过滤方式是移不变的, 上述引入的高斯范围滤波器对图像强度的整体加性变化不敏感, 因此

无偏: 如果过滤器对 $(f(\xi), x(\xi))$ 的处理结果是 $(h(\xi), x(\xi))$, 那么同样的滤波器应用于 $(f(\xi), x(\xi))$ 也会得到 $(h(\xi), x(\xi))$, 因为 $(f(\xi), x(\xi))$ 和 $(h(\xi), x(\xi))$ 都是 $(f(\xi), x(\xi))$ 的线性变换。当然, 范围滤波器也是移不变的, 这一点很容易从表达式 (3) 和 (4) 中验证。

3 范围滤波与双边滤波比较

在上一节中, 我们将范围滤波与域滤波结合以生成双边滤波器。我们现在展示这种组合是必不可少的。为了简化记号, 我们仅讨论黑白图像, 但类似的结果也适用于多频带图像。本节的主要观点是, 单独使用范围滤波只会修改其应用图像的灰度图。这是由于范围滤波器没有空间概念这一事实所直接导致的。

令 $\nu(\xi)$ 表示输入图像灰度级的频率分布。在离散情况下, $\nu(\xi)$ 是灰度直方图: $\nu(\xi)$ 通常是介于 $\nu(\xi)$ 和 $\nu(\xi)$ 之间的整数, 而 $\nu(\xi)$ 表示具有灰度值为 $\nu(\xi)$ 的图像像素的比例。在连续情况下, $\nu(\xi)$ 表示灰度值介于 $\nu(\xi)$ 和 $\nu(\xi)$ 之间的图像区域比例。为了保持符号一致性, 我们继续讨论连续情况, 如同前一节所述。

简单的推导 (由于篇幅限制省略) 表明, 范围过滤器的表达式 (3) 和 (4) 可以合并为以下形式:

$$\nu = \int_0^{\infty} \phi \nu(\phi) q(\phi) d\phi \quad (7)$$

其中

$$\nu(\phi, \xi) = \int_0^{\infty} s(\phi, \xi) \nu(\phi) q(\phi) d\phi$$

不论位置 $\mathbf{x}$ 如何, 式 (7) 展示了范围滤波是一种灰度级的简单变换。映射核 $\nu(\phi, \xi)$ 是一个密度函数, 即它是非负的, 并且其积分值为1。它等于以 ξ 为中心、由相似性函数 s 加权后的直方图 $\nu(\phi)$ 并归一化到单位面积。由于 ν 是

formally a density function, equation (7) represents a mean. We can therefore conclude with the following result:

Range filtering merely transforms the gray map of the input image. The transformed gray value is equal to the mean of the input’s histogram values around the input gray level f , weighted by the range similarity function s centered at f .

It is useful to analyze the nature of this gray map transformation in view of our discussion of bilateral filtering. Specifically, we want to show that

Range filtering compresses unimodal histograms.

In fact, suppose that the histogram $\nu(f)$ of the input image is a single-mode curve as in figure 2 (a), and consider an input value of f located on either side of this bell curve. Since the symmetric similarity function s is centered at f , on the rising flank of the histogram, the product $s\nu$ produces a skewed density $\tau(\phi, f)$. On the left side of the bell τ is skewed to the right, and vice versa. Since the transformed value h is the mean of this skewed density, we have $h > f$ on the left side and $h < f$ on the right side. Thus, the flanks of the histogram are compressed together.

At first, the result that range filtering is a simple remapping of the gray map seems to make range filtering rather useless. Things are very different, however, when range filtering is combined with domain filtering to yield *bilateral* filtering, as shown in equations (5) and (6). In fact, consider first a domain closeness function c that is constant within a window centered at x , and is zero elsewhere. Then, the bilateral filter is simply a range filter applied to the window. The filtered image is still the result of a local remapping of the gray map, but a very interesting one, because the remapping is different at different points in the image.

For instance, the solid curve in figure 2 (b) shows the histogram of the step image of figure 1 (a). This histogram is bimodal, and its two lobes are sufficiently separate to allow us to apply the compression result above to each lobe. The dashed line in figure 2 (b) shows the effect of bilateral filtering on the histogram. The compression effect is obvious, and corresponds to the separate smoothing of the light and dark sides, shown in figure 1 (c). Similar considerations apply when the closeness function has a profile other than constant, as for instance the Gaussian profile shown in section 2, which emphasizes points that are closer to the center of the window.

4 Experiments with Black-and-White Images

In this section we analyze the performance of bilateral filters on black-and-white images. Figure 5 (a) and 5 (b) in the color plates show the potential of bilateral filtering for

the removal of texture. Some amount of gray-level quantization can be seen in figure 5 (b), but this is caused by the printing process, not by the filter. The picture “simplification” illustrated by figure 5 (b) can be useful for data reduction without loss of overall shape features in applications such as image transmission, picture editing and manipulation, image description for retrieval. Notice that the kitten’s whiskers, much thinner than the filter’s window, remain crisp after filtering. The intensity values of dark pixels are averaged together from both sides of the whisker, while the bright pixels from the whisker itself are ignored because of the range component of the filter. Conversely, when the filter is centered somewhere on a whisker, only whisker pixel values are averaged together.

Figure 3 shows the effect of different values of the parameters σ_d and σ_r on the resulting image. Rows correspond to different amounts of domain filtering, columns to different amounts of range filtering. When the value of the range filtering constant σ_r is large (100 or 300) with respect to the overall range of values in the image (1 through 254), the range component of the filter has little effect for small σ_d : all pixel values in any given neighborhood have about the same weight from range filtering, and the domain filter acts as a standard Gaussian filter. This effect can be seen in the last two columns of figure (3). For smaller values of the range filter parameter σ_r (10 or 30), range filtering dominates perceptually because it preserves edges.

However, for $\sigma_d = 10$, image detail that was removed by smaller values of σ_d reappears. This apparently paradoxical effect can be noticed in the last row of figure 3, and in particularly dramatic form for $\sigma_r = 100$, $\sigma_d = 10$. This image is crisper than that above it, although somewhat hazy. This is a consequence of the gray map transformation and histogram compression results discussed in section 3. In fact, $\sigma_d = 10$ is a very broad Gaussian, and the bilateral filter becomes essentially a range filter. Since intensity values are simply remapped by a range filter, no loss of detail occurs. Furthermore, since a range filter compresses the image histogram, the output image appears to be hazy. Figure 2 (c) shows the histograms for the input image and for the two output images for $\sigma_r = 100$, $\sigma_d = 3$, and for $\sigma_r = 100$, $\sigma_d = 10$. The compression effect is obvious.

Bilateral filtering with parameters $\sigma_d = 3$ pixels and $\sigma_r = 50$ intensity values is applied to the image in figure 4 (a) to yield the image in figure 4 (b). Notice that most of the fine texture has been filtered away, and yet all contours are as crisp as in the original image.

Figure 4 (c) shows a detail of figure 4 (a), and figure 4 (d) shows the corresponding filtered version. The two onions have assumed a graphics-like appearance, and the fine texture has gone. However, the overall shading is preserved, because it is well within the band of the domain

形式上是一个密度函数，方程（7）表示一个平均值。因此，我们可以得出以下结论：

范围滤波仅将输入图像的灰度图进行变换。
变换后的灰度值等于以输入灰度级为中心的直方图值在该中心点附近的加权平均，权重由围绕该中心点的范围相似函数给出。

从我们对双边滤波器的讨论来看，分析这种灰度映射变换的性质是有用的。具体来说，我们希望证明的是

范围过滤可以压缩单模直方图。

事实上，假设输入图像的直方图 $N(\lambda)$ 如图2（a）所示为单峰曲线，并考虑位于该钟形曲线两侧的输入值 λ_0 。由于对称相似函数 $\rho(\lambda, \lambda_0)$ 以 λ_0 为中心，在直方图上升沿处，乘积 $\rho(\lambda, \lambda_0) N(\lambda)$ 会产生偏斜密度 $\lambda(\rho, \lambda_0)$ 。在钟形曲线左侧的部分向右偏斜，反之亦然。因为变换后的值 λ_p 是该偏斜密度的均值，所以在左侧我们有 $\lambda_p > \lambda_0$ ，而在右侧则为 $\lambda_p < \lambda_0$ 。因此，直方图的边缘被压缩在一起。

最初，结果表明范围滤波只是灰度映射的简单重映射，这似乎使范围滤波显得无足轻重。然而，当将范围滤波与域滤波结合使用以生成bilateral滤波时（如方程(5)和(6)所示），情况就大不相同了。实际上，首先考虑一个在x为中心的窗口内为常数且其他地方为零的域接近度函数 c_x 。此时，双边滤波器只是对窗口应用范围滤波的结果。滤波后的图像仍然是局部灰度映射重映射的结果，但这是一个非常有趣的重映射，因为不同图像点处的重映射是不同的。

例如，图2(b)中的实线显示了图1(a)中步进图像的直方图。该直方图是双峰型，并且其两个峰值足够分离，使得我们可以将上述压缩结果分别应用于每个峰值区域。图2(b)中的虚线展示了双边滤波对直方图的影响。这种压缩效果很明显，对应于图1(c)中亮部和暗部的独立平滑处理。当接近度函数具有非恒定轮廓（例如第2节所示的高斯轮廓），该轮廓强调靠近窗口中心的点时，类似的考虑同样适用。

4 黑白图像实验

在本节中，我们分析了双边滤波器在黑白图像上的性能。彩色插图中的图5（a）和图5（b）展示了双边滤波的潜力。

去除纹理。图5(b)中可以看到一定程度的灰度量化，但这由打印过程引起，并非滤波器所致。如图5(b)所示的图像“简化”在诸如图像传输、图片编辑和处理、图像描述以供检索等应用中可以用于数据缩减而不丢失整体形状特征。请注意，猫胡须比滤波器窗口细得多，在经过滤波后仍然保持清晰。胡须两侧的暗像素强度值被平均化，而来自胡须本身的亮像素由于滤波器范围组件的作用被忽略。相反地，当滤波器中心位于某根胡须上时，只有胡须上的像素值会被平均化在一起。

图3展示了参数 σ_s 和 σ_r 的不同取值对生成图像的影响。行对应不同的域过滤量，列对应不同的范围过滤量。当范围过滤常数 σ_r 相对于图像中的整体值范围（从1到254）较大（如100或300）时，在较小的 σ_s 下，范围滤波器的作用微乎其微：在任何给定邻域内，所有像素值都具有大致相同的权重，此时域过滤器充当标准高斯滤波器。这一效果可以在图(3)的最后一列中观察到。对于较小的范围滤波参数 σ_r （如10或30），范围过滤在感知上占主导地位，因为它能保留边缘。

然而，对于 $\sigma_r = 10$ ，较小的 σ_s 值所移除的图像细节重新出现。这种看似矛盾的效果可以在图3的最后一行中观察到，并且在 $\sigma_r = 100$ 和 $\sigma_r = 10$ 中尤为明显。该图像比其上方的图像更清晰，尽管略显模糊。这是灰度映射变换以及第3节讨论的直方图压缩结果所导致的后果。实际上， $\sigma_r = 10$ 是一个非常宽广的高斯分布，双边滤波器基本上变成了范围滤波器。由于强度值只是通过范围滤波器重新映射，因此不会丢失细节。此外，由于范围滤波器会压缩图像直方图，输出图像看起来略显模糊。图2（c）显示了输入图像以及对于 $\sigma_r = 100$ 和 $\sigma_r = 3$ ， $\sigma_s = 100$ 和 $\sigma_s = 10$ 的两个输出图像的直方图。压缩效果非常明显。

双边滤波器采用参数 $\sigma_r = 3$ 像素和 $\sigma_s = 30$ 强度值应用于图4(a)中的图像，得到图4(b)所示的图像。请注意，大部分精细纹理已被过滤掉，而所有轮廓依然像原始图像一样清晰锐利。

图4(c)展示了图4(a)的细节，而图4(d)则显示了对应的滤波版本。两个洋葱呈现出类似图形的效果，细腻的纹理已经消失。然而，整体的阴影效果得以保留，因为这完全处于域内的频带范围内。

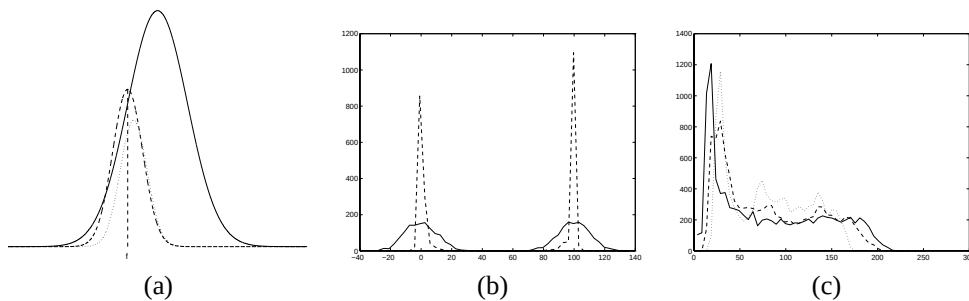


Figure 2: (a) A unimodal image histogram ν (solid), and the Gaussian similarity function s (dashed). Their normalized product τ (dotted) is skewed to the right. (b) Histogram (solid) of image intensities for the step in figure 1 (a) and (dashed) for the filtered image in figure 1 (c). (c) Histogram of image intensities for the image in figure 5 (a) (solid) and for the output images with $\sigma_r = 100$, $\sigma_d = 3$ (dashed) and with $\sigma_r = 100$, $\sigma_d = 10$ (dotted) from figure 3.

filter and is almost unaffected by the range filter. Also, the boundaries of the onions are preserved.

In terms of computational cost, the bilateral filter is twice as expensive as a nonseparable domain filter of the same size. The range component depends nonlinearly on the image, and is nonseparable. A simple trick that decreases computation cost considerably is to precompute all values for the similarity function $s(\phi, f)$. In the Gaussian case, if the image has n levels, there are $2n + 1$ possible values for s , one for each possible value of the difference $\phi - f$.

5 Experiments with Color Images

For black-and-white images, intensities between any two grey levels are still grey levels. As a consequence, when smoothing black-and-white images with a standard low-pass filter, intermediate levels of gray are produced across edges, thereby producing blurred images. With color images, an additional complication arises from the fact that between any two colors there are other, often rather different colors. For instance, between blue and red there are various shades of pink and purple. Thus, disturbing color bands may be produced when smoothing across color edges. The smoothed image does not just look blurred, it also exhibits odd-looking, colored auras around objects. Figure 6 (a) in the color plates shows a detail from a picture with a red jacket against a blue sky. Even in this unblurred picture, a thin pink-purple line is visible, and is caused by a combination of lens blurring and pixel averaging. In fact, pixels along the boundary, when projected back into the scene, intersect both red jacket and blue sky, and the resulting color is the pink average of red and blue. When smoothing, this effect is emphasized, as the broad, blurred pink-purple area in figure 6 (b) shows.

To address this difficulty, edge-preserving smoothing could be applied to the red, green, and blue components of the image separately. However, the intensity profiles across the edge in the three color bands are in general different. Separate smoothing results in an even more pronounced

pink-purple band than in the original, as shown in figure 6 (c). The pink-purple band, however, is not widened as it is in the standard-blurred version of figure 6 (b).

A much better result can be obtained with bilateral filtering. In fact, a bilateral filter allows combining the three color bands appropriately, and measuring photometric distances between pixels in the combined space. Moreover, this combined distance can be made to correspond closely to perceived dissimilarity by using Euclidean distance in the CIE-Lab color space [16]. This space is based on a large body of psychophysical data concerning color-matching experiments performed by human observers. In this space, small Euclidean distances correlate strongly with the perception of color discrepancy as experienced by an “average” color-normal human observer. Thus, in a sense, bilateral filtering performed in the CIE-Lab color space is the most natural type of filtering for color images: only perceptually similar colors are averaged together, and only perceptually important edges are preserved. Figure 6 (d) shows the image resulting from bilateral smoothing of the image in figure 6 (a). The pink band has shrunk considerably, and no extraneous colors appear.

Figure 7 (c) in the color plates shows the result of five iterations of bilateral filtering of the image in figure 7 (a). While a single iteration produces a much cleaner image (figure 7 (b)) than the original, and is probably sufficient for most image processing needs, multiple iterations have the effect of flattening the colors in an image considerably, but without blurring edges. The resulting image has a much smaller color map, and the effects of bilateral filtering are easier to see when displayed on a printed page. Notice the cartoon-like appearance of figure 7 (c). All shadows and edges are preserved, but most of the shading is gone, and no “new” colors are introduced by filtering.

6 Relations with Previous Work

The literature on edge-preserving filtering is vast, and we make no attempt to summarize it. An early survey can

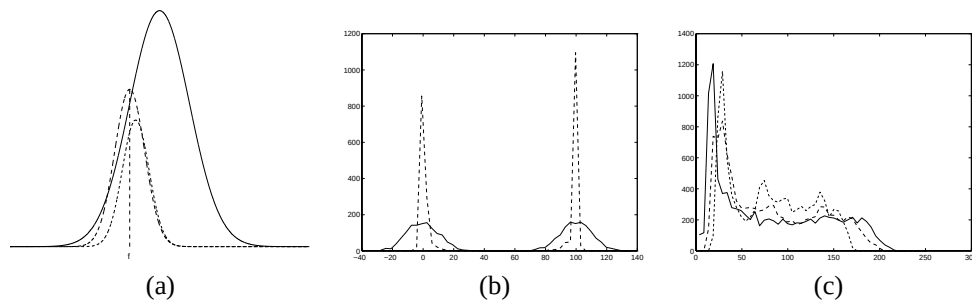


图2: (a) 单峰图像直方图 (实线), 以及高斯相似函数 (虚线)。它们的归一化乘积 (点划线) 向右偏斜。(b) 图1(a) 中阶梯图像的强度直方图 (实线) 和图1(c) 中滤波后图像的强度直方图 (虚线)。(c) 图5(a) 中图像的强度直方图 (实线), 以及来自图3中的输出图像 $q_u = 100, q_v = 3$ (虚线) 和 $q_u = 100, q_v = 10$ (点划线) 的强度直方图。

过滤器对其影响很小, 几乎不受范围过滤器的影响。此外, 洋葱的边界得以保留。

在计算成本方面, 双边滤波器的开销是同尺寸非可分离域滤波器的两倍。范围组件与图像呈非线性关系, 并且是非可分离的。一个可以显著降低计算成本的小技巧是在预处理阶段计算所有相似函数 $\phi(\phi_i, \phi_j)$ 的值。在高斯情况下, 如果图像具有 u 个灰度级, 则对于每个可能的差值 $\phi_i - \phi_j$, 就有 $2u + 1$ 种可能的取值。

5 实验与彩色图像

对于黑白图像, 任意两个灰度级别之间的强度仍然是灰度级别。因此, 当使用标准低通滤波器平滑处理黑白图像时, 在边缘处会产生不同程度的灰色过渡, 从而导致图像模糊。而对于彩色图像, 则会因为任意两种颜色之间存在其他往往差异较大的颜色而产生额外复杂性。例如, 在蓝色和红色之间有各种粉红和紫色的色调。因此, 在跨色彩边界进行平滑处理时, 可能会生成令人不安的颜色带。经过平滑处理后的图像不仅看起来模糊不清, 还会在物体周围出现奇怪的彩色光环。彩图6(a) 展示了一张细节图片, 其中一件红色夹克位于蓝色天空背景下。即使在这幅未被模糊的图片中, 也能看到一条细薄的粉紫色线条, 这是镜头模糊和像素平均化共同作用的结果。实际上, 在边界上的像素投影回场景时会同时穿过红色夹克和蓝色天空区域, 因此产生的颜色是红蓝混合后的粉色平均值。在进行平滑处理时, 这种效果会被放大, 如图6(b) 所示的宽广、模糊的粉紫色区域所展示。

为了应对这一难题, 可以分别对图像红色、绿色和蓝色组件应用边缘保持平滑处理。然而, 三个色彩通道中的边缘强度分布通常各不相同。单独进行平滑处理会导致效果更加明显地偏离预期。

与原图相比, 如图6(c) 所示, 出现了粉紫色带, 但是该粉紫色带并未像图6(b) 的标准模糊版本那样变宽。

通过双边滤波可以得到更好的结果。事实上, 双边滤波允许适当组合三个颜色通道, 并在组合空间中测量像素之间的光度距离。此外, 通过在CIE-Lab色彩空间中使用欧几里得距离[16], 这种组合距离可以与感知到的差异紧密对应。该空间基于大量关于人类观察者进行的颜色匹配实验的心理物理数据。在这个空间中, 较小的欧氏距离与“平均”正常色觉的人类观察者的颜色差异感知密切相关。因此, 在某种意义上, CIE-Lab色彩空间中的双边滤波是最自然的颜色图像滤波类型: 只有感知相似的颜色才会被平均化, 而重要的边缘则会被保留下来。图6(d) 显示了对图6(a) 进行双边平滑处理后的图像结果, 粉红色带明显缩小, 并且没有出现额外的颜色。

图7(c) 中的彩色插图展示了对图7(a) 中图像进行五次双边滤波后的结果。虽然单次迭代就能生成比原图更干净的图像 (如图7(b) 所示), 对于大多数图像处理需求来说这已经足够了, 但多次迭代会使图像颜色变得相当平坦, 而不会模糊边缘。最终得到的图像具有更小的颜色映射表, 在打印页面上显示时双边滤波的效果更容易观察到。请注意图7(c) 呈现出类似卡通画的外观: 所有阴影和边缘都得到了保留, 但大部分渐变效果消失了, 并且过滤过程中没有引入新的颜色。

6 相关工作关系

边缘保持滤波的文献非常丰富, 我们不试图对其进行总结。早期的一篇综述可以参考……

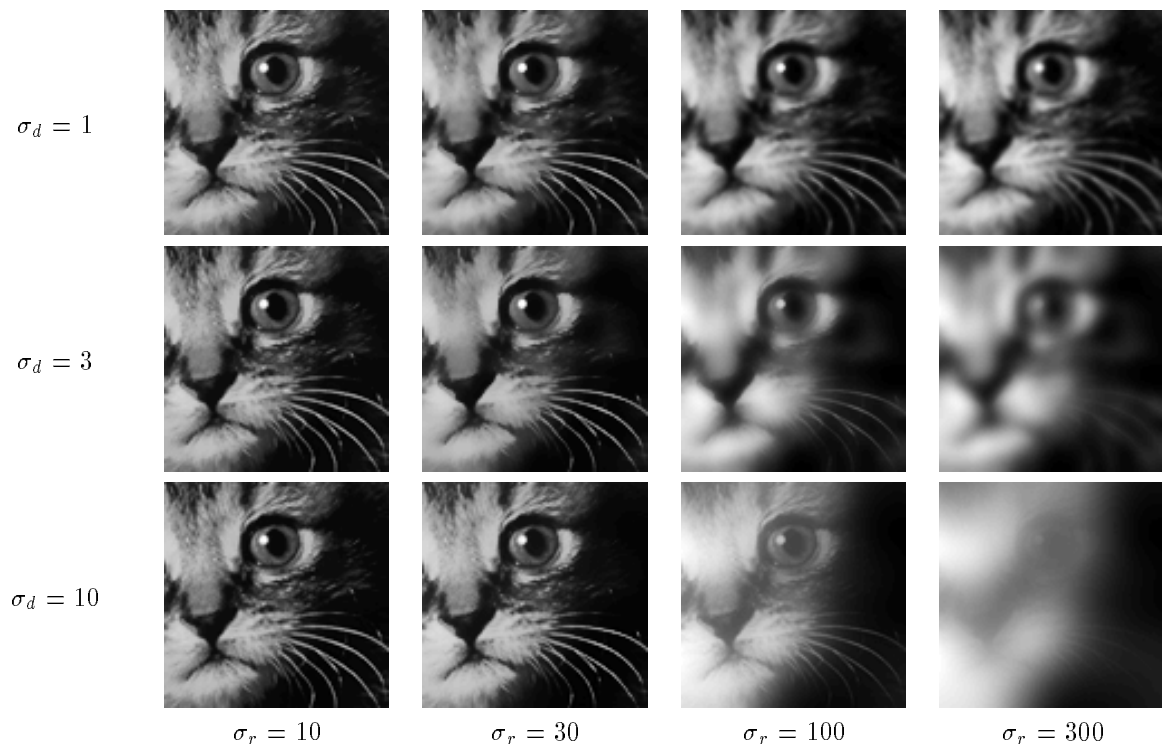


Figure 3: A detail from figure 5 (a) processed with bilateral filters with various range and domain parameter values.

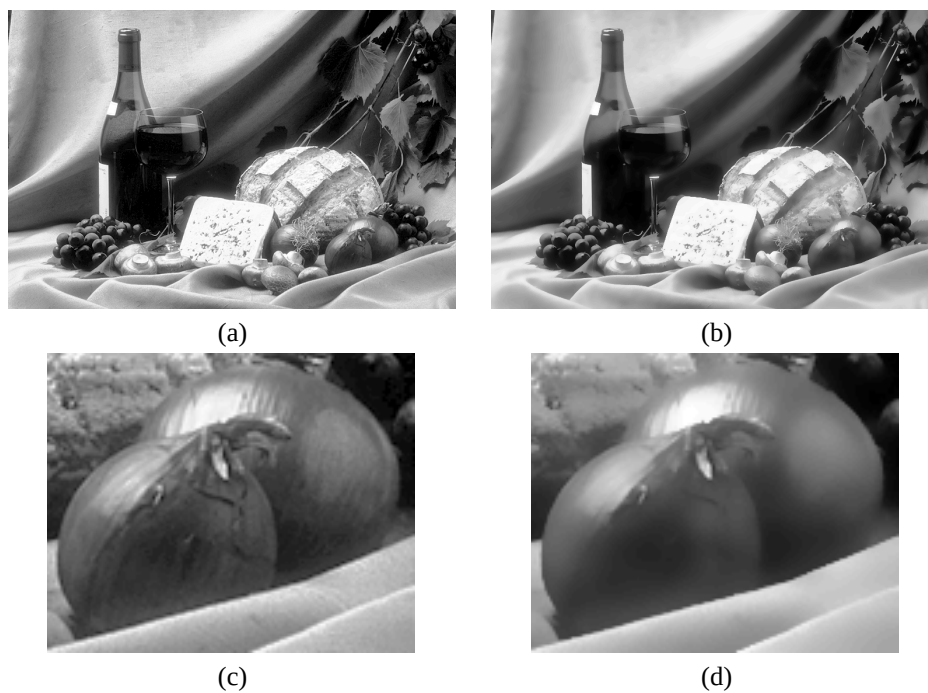


Figure 4: A picture before (a) and after (b) bilateral filtering. (c,d) are details from (a,b).

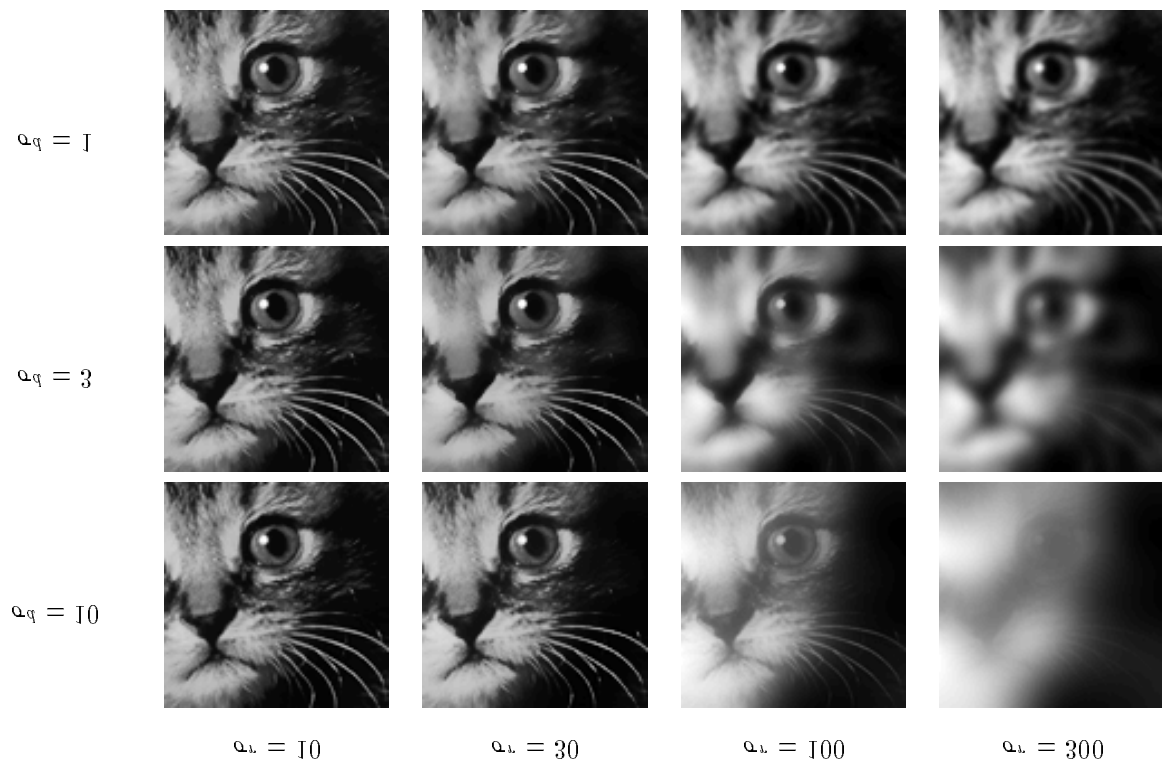


图3: 细节展示 图5 (a) 经过了不同范围和域参数的双边滤波处理。 计量值。

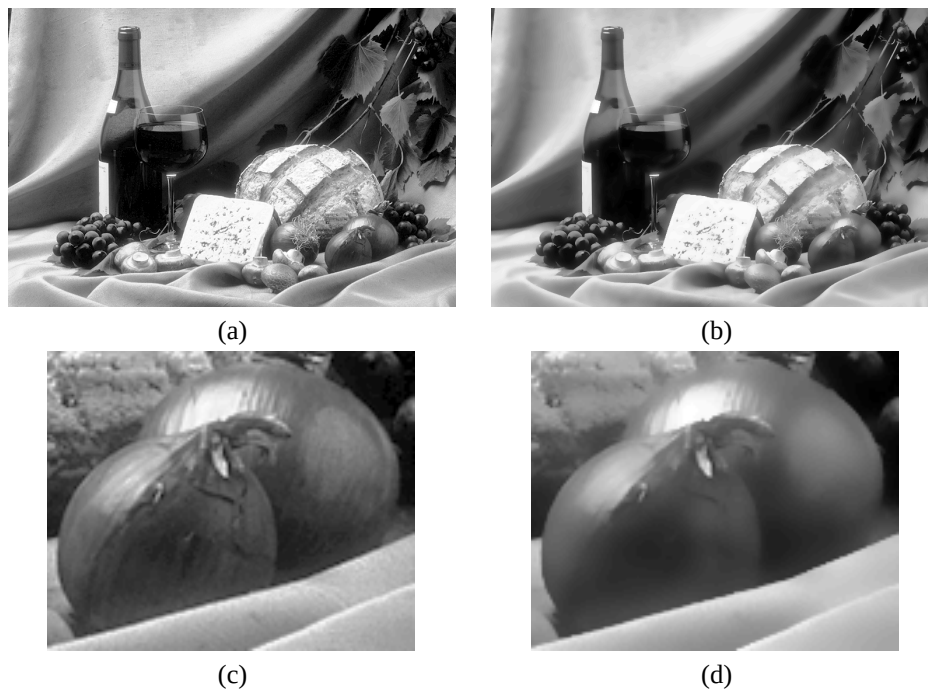


图4: 双边滤波前 (a) 和后的图像 (b)。(c,d)分别为(a,b)的细节放大。

be found in [8], quantitative comparisons in [2], and more recent results in [1]. In the latter paper, the notion that neighboring pixels should be averaged only when they are similar enough to the central pixels is incorporated into the definition of the so-called “G-neighbors.” Thus, G-neighbors are in a sense an extreme case of our method, in which a pixel is either counted or it is not. Neighbors in [1] are strictly adjacent pixels, so iteration is necessary.

A common technique for preserving edges during smoothing is to compute the median in the filter’s support, rather than the mean. Examples of this approach are [6, 9], and an important variation [3] that uses K -means instead of medians to achieve greater robustness.

More related to our approach are weighting schemes that essentially average values within a sliding window, but change the weights according to local differential [4, 15] or statistical [10, 7] measures. Of these, the most closely related article is [10], which contains the idea of multiplying a geometric and a photometric term in the filter kernel. However, that paper uses rational functions of distance as weights, with a consequent slow decay rate. This forces application of the filter to only the immediate neighbors of every pixel, and mandates multiple iterations of the filter. In contrast, our bilateral filter uses Gaussians as a way to enforce what Overton and Weimouth call “center pixel dominance.” A single iteration drastically “cleans” an image of noise and other small fluctuations, and preserves edges even when a very wide Gaussian is used for the domain component. Multiple iterations are still useful in some circumstances, as illustrated in figure 7 (c), but only when a cartoon-like image is desired as the output. In addition, no metrics are proposed in [10] (or in any of the other papers mentioned above) for color images, and no analysis is given of the interaction between the range and the domain components. Our discussions in sections 3 and 5 address both these issues in substantial detail.

7 Conclusions

In this paper we have introduced the concept of bilateral filtering for edge-preserving smoothing. The generality of bilateral filtering is analogous to that of traditional filtering, which we called domain filtering in this paper. The explicit enforcement of a photometric distance in the range component of a bilateral filter makes it possible to process color images in a perceptually appropriate fashion.

The parameters used for bilateral filtering in our illustrative examples were to some extent arbitrary. This is however a consequence of the generality of this technique. In fact, just as the parameters of domain filters depend on image properties and on the intended result, so do those of bilateral filters. Given a specific application, techniques for the automatic design of filter profiles and parameter values may be possible.

Also, analogously to what happens for domain filtering, similarity metrics different from Gaussian can be defined for bilateral filtering as well. In addition, range filters can be combined with different types of domain filters, including oriented filters. Perhaps even a new scale space can be defined in which the range filter parameter σ_r corresponds to scale. In such a space, detail is lost for increasing σ_r , but edges are preserved at all range scales that are below the maximum image intensity value. Although bilateral filters are harder to analyze than domain filters, because of their nonlinear nature, we hope that other researchers will find them as intriguing as they are to us, and will contribute to their understanding.

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见文献[8], 定量比较参见文献[2], 最近的结果可参考文献[1]。在后者的论文中, 相邻像素仅当它们与中心像素足够相似时才进行平均的概念被纳入所谓的“G-邻域”的定义之中。因此, “G-邻域”在某种程度上是我们方法的一个极端情况, 在这种情况下, 一个像素要么被计数, 要么不被计数。文献[1]中的邻居严格指相邻的像素, 因此需要迭代处理。

在平滑处理过程中保留边缘的一种常见技术是计算滤波器支撑区域内的中位数, 而不是平均值。这种方法的实例见文献[6, 9], 以及一个重要的变体[3], 该变体使用 V 均值而非中位数来实现更高的鲁棒性。

更相关于我们的方法的是那些本质上在滑动窗口内求值平均的加权方案, 但会根据局部微分[4,15]或统计[10,7]度量来改变权重。其中最密切相关的一篇文章是[10], 该文章提出了在滤波器核中乘以几何项和光度项的思想。然而, 该论文使用距离的有理函数作为权重, 导致衰减速率较慢。这迫使滤波器仅应用于每个像素的直接邻域, 并且需要多次迭代滤波器。相比之下, 我们的双边滤波器采用高斯分布来强制执行Overton和Weimouth所说的“中心像素主导”。单次迭代即可显著清除图像中的噪声和其他小波动, 并在使用非常宽的高斯分布作为领域分量时仍能保持边缘。在某些情况下多次迭代仍然有用, 如图7(c)所示, 但仅当希望输出类似卡通的效果时才适用。此外, 在[10] (以及其他上述论文) 中没有为彩色图像提出任何度量标准, 并且未分析范围和领域组件之间的相互作用。我们在第3节和第5节中的讨论详细地解决了这两个问题。

7 结论

在本文中, 我们介绍了双边滤波的概念, 用于保持边缘的平滑处理。双边滤波的通用性与传统的域滤波(我们在本文中称之为域滤波)类似。双边滤波器范围组件中明确实施的光度距离强制执行使得以感知上适当的方式处理彩色图像成为可能。

用于我们示例中的双边滤波参数在一定程度上是任意选择的。然而, 这是该技术通用性的结果。事实上, 正如领域滤波器的参数取决于图像属性和预期效果一样, 双边滤波器的参数也是如此。对于特定应用而言, 自动设计滤波器配置文件和技术参数的方法可能是可行的。

同样地, 类似于领域过滤的情况, 双边滤波也可以定义不同于高斯的相似性度量。此外, 范围滤波器可以与不同类型的空间滤波器结合使用, 包括定向滤波器。或许甚至可以在一个新的尺度空间中定义范围滤波器参数 α_v 作为尺度。在这种空间中, 随着 α_v 的增加会丢失细节, 但在低于最大图像强度值的所有范围内边缘得以保留。尽管双边滤波器由于其非线性特性难以分析, 我们希望其他研究人员也会像我们一样对它们感到好奇, 并为理解它们做出贡献。

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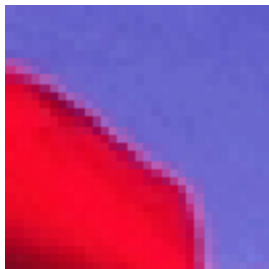


(a)

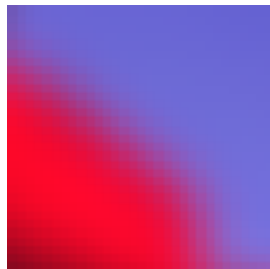


(b)

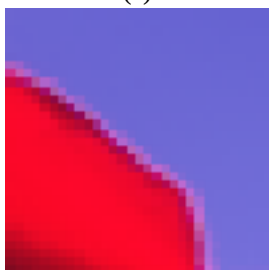
Figure 5: A picture before (a) and after (b) bilateral filtering.



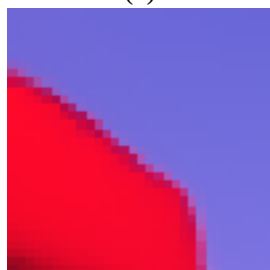
(a)



(b)



(c)



(d)



(a)



(b)



(c)

Figure 7: **[above]** (a) A color image, and its bilaterally smoothed versions after one (b) and five (c) iterations.

Figure 6: **[left]** (a) A detail from a picture with a red jacket against a blue sky. The thin, pink line in (a) is spread and blurred by ordinary low-pass filtering (b). Separate bilateral filtering (c) of the red, green, blue components sharpens the pink band, but does not shrink it. Combined bilateral filtering (d) in CIE-Lab color space shrinks the pink band, and introduces no spurious colors.

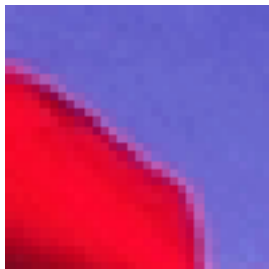


(a)

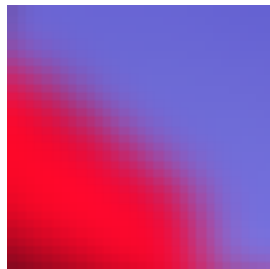


(b)

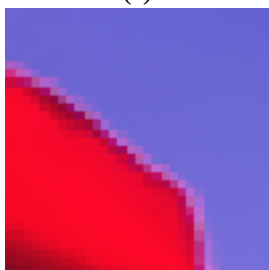
图5: 双边滤波前 (a) 和双边滤波后 (b) 的图片。



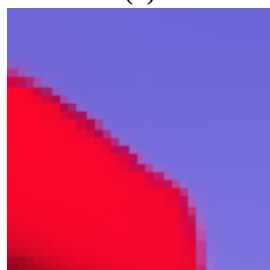
(a)



(b)



(c)



(d)



(a)



(b)



(c)

图7: [上方] (a) 一幅彩色图像及其经过一次 (b) 和五次 (c) 双边滤波后的版本。

图6: [左] (a) 一张穿着红色夹克的人在蓝天背景下的细节照片。普通低通滤波器处理后, (a) 中的细粉色线条变得扩散和模糊 (b)。分别对红、绿、蓝分量进行双边滤波处理 (c), 可以锐化粉红线带, 但不会使其缩小。而在CIE-Lab颜色空间中结合使用双边滤波处理 (d), 则可以使粉红线带缩小, 并且不引入任何伪彩色。